

# PAUL KOIMBURI KAHURA

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| <https://paulkahura.github.io/portfolio>

## PROFESSIONAL SUMMARY

Innovative Software Engineer and Serial Entrepreneur with extensive experience architecting scalable, full-stack solutions. Co-founded/ architected 3 startups, successfully bringing 2 to market with international reach. Expert in the **MERN stack**, **Python**, and **Cloud Infrastructure (AWS/GCP)**, with a specialized focus on **AI integration**, **Data Pipelines**, and **Secure Multi-tenant Architectures**. Proven ability to bridge the gap between complex backend logic and accessible user experiences for diverse populations.

## TECHNICAL SKILLS

- **Backend:** Node.js, TypeScript, Python (Django/Flask/FastAPI), GraphQL, REST APIs.
- **Frontend:** React.js, Angular, Vue.js, Flutter, Tailwind CSS.
- **Data & Cloud:** AWS (Redshift, Lambda, QuickSight), dbt, MongoDB, PostgreSQL, Firebase, Redis, Edge Computing.
- **AI & ML:** LLM Integration (OpenAI, Gemini etc), Computer Vision, NLP Workflows, RAG Pipelines.
- **DevOps:** Docker, CI/CD, Git, Nginx, Azure, Cloudformation.
- **Mobile:** Flutter, Dart, Firebase, Riverpod, MapLibre, Firestore.

## WORK EXPERIENCE

**Elewa / Italanta** | *Full Stack Software Engineer Nairobi, Kenya* | Aug 2023 – Present

- Collaborated in a dynamic startup to build AI-driven educational tools using **Node.js**, **Python**, **TypeScript**, **Flutter**, and **GraphQL**.
- **AI Integration/ MCP:** Integrated multiple AI models (Gemini, OpenAI, Anthropic, ) to create intelligent chatbot and IVR systems accessible to users with limited internet access. Also worked on internal mcps that helped the company in their day to day work.
- **FarmBetter AI IVR System:** Designed an IVR solution integrating **Azure TTS**, **Pinecone**, **Twilio**, and **Google Cloud Functions** to deliver personalized agricultural advice to farmers via phone.

- **AI Crisis Communication System** : Architected a multi-tenant crisis management platform leveraging AI for sentiment analysis, real-time updates, and automated response workflows.
- **Scale**: Supported applications with over 10,000 users by resolving technical issues and optimizing system uptime.

### **Shockwolves BV(under Elewa)**

Nairobi, Kenya | Jan 2026 – Present

Led a team to build AI-driven educational tools and enterprise data platforms using Node.js, TypeScript, Flutter, GraphQL, Python, and AWS.

### **TIVTAV — Secure Multi-Tenant Document Analytics Platform (Healthcare/Dental):**

- **Full-Silo Multi-Tenant Architecture**: Designed and implemented a strict full-silo tenancy model on AWS where each tenant receives a fully dedicated VPC, Redshift Serverless namespace, S3 buckets, Lambda functions, Cognito User Pool, and Secrets Manager entries ensuring zero cross-tenant data paths at both the network and application layer.
- **Infrastructure as Code**: Authored the entire cloud infrastructure using AWS CDK (Python 3.13), deploying ~50 AWS resources per tenant via automated CloudFormation stacks from a versioned S3 template.
- **Automated Provisioning Pipeline**: Built a self-healing onboarding pipeline using API Gateway, AWS Step Functions, Lambda, EventBridge, and DynamoDB to provision fully isolated tenant environments end-to-end (including Redshift schema initialisation and QuickSight setup) in under 15 minutes.
- **AI Document Extraction Pipeline**: Integrated ExtendAI to ingest and extract structured data from unstructured invoices and payslips; Lambda functions handle webhook signature verification, workflow triggering, and S3-based event ingestion into Redshift via Zero-ETL S3 auto-copy jobs.
- **Private Networking & Security**: Configured all tenant VPCs with PRIVATE\_ISOLATED subnets and AWS PrivateLink endpoints only no NAT gateways, no internet egress eliminating data exfiltration attack vectors. Per-tenant IAM roles, scoped Secrets Manager policies, and credential isolation per tenant enforced at every layer.
- **CI/CD**: Set up GitHub Actions workflows with OIDC-authenticated ECR image publishing and automated CDK deployment pipelines.

**Three Point Devhub** | *Lead Software Engineer Nairobi, Kenya | tech startup consultancy focused on building market-ready SaaS products. Led full-cycle delivery from ideation to deployment.*

- **Retail Data Intelligence Platform (SaaS):**
  - **Architecture**: Architected a secure, **siloed multi-tenant** system on AWS to provide strict data isolation for retail chains and store owners.

- **Data Pipeline:** Built an end-to-end ELT pipeline using **Python**, **dbt**, and **Amazon Redshift**. Automated the ingestion of unstructured documents (invoices, payslips) using **Extend AI**, transforming them into structured, queryable data.
- **Visualization:** Developed a **React** and **GraphQL** frontend integrated with **Amazon QuickSight** to provide real-time financial dashboards and cross-store performance metrics.
- **Blue Box AI (Surveillance Analytics):**
  - **Edge-Cloud Hybrid:** Designed a system that combines **Edge Computing** with Cloud AI to process CCTV footage efficiently, reducing latency and bandwidth costs.
  - **Natural Language Interface:** Integrated **Computer Vision** with **LLMs** to allow users to query video feeds via natural language on **WhatsApp** (e.g., "Show me who entered the store at 10 AM"), democratizing security insights for non-technical users.
- **Zuka Safari**

Built a real-time psv tracking and mobility platform for Nairobi with separate consumer and driver applications. Developed features for live vehicle location tracking, route discovery, fare visibility, fuel station availability, and trip management, using Firebase Authentication and Firestore for real-time data sync. Designed the app as an aggregator platform with adapter-based integrations for external fleet providers and CCTV occupancy systems(using line tracking), making it possible to normalize vehicle, route, and passenger data from multiple sources into a unified mobile experience.

#### **Netbot Solutions** | *Full Stack Software Engineer Nairobi, Kenya* | Oct 2022 – Jul 2023

- **Property Management System:** Designed and implemented a comprehensive visitor management system using **Django** and **REST APIs**, integrating with **Flutter** mobile apps.
- **Data Security:** Managed database schemas and implemented robust authentication and access control systems for tenants and building managers.
- **Analytics:** Built a data reporting dashboard to visualize occupancy rates and maintenance costs, aiding decision-making for property owners.
- **Agile Collaboration:** Utilized Jira and Notion for sprint planning and performance tracking.

#### **Novatta Africa** | *Web Developer for startup Nairobi, Kenya* | 2019 – 2024

- Built and maintained multiple client-facing applications including sports betting platforms and e-commerce sites.
- Developed high-performance user interfaces with a focus on accessibility and design consistency.
- Collaborated with cross-functional teams to integrate payment gateways and third-party APIs into cloud-hosted infrastructure.

## EDUCATION

**BSc Mathematics and Computer Science** Jomo Kenyatta University of Agriculture and Technology (JKUAT)

## CERTIFICATIONS

- **Software Development for Cloud Computing** (Coursera)
- **EF SET English certificate (C2 Proficient)**

## RELEVANT PROJECTS & VOLUNTEER WORK

- **Volunteer Software Engineer:** I have developed multiple custom web solutions for local businesses, churches and children's organisations to digitize operations and improve online presence.
- **Open Source Contributor:** Passionate about contributing to projects that make technology more accessible and inclusive